

Causal Operator–Operand Factorization in the Residual Stream of LLMs

Matias Podeley

Independent researcher

mpodeley@gmail.com

Abstract

How do language models represent a relational operation — *currency-of*, *capital-of* — as distinct from the entity it applies to? Across Qwen3-1.7B/8B and Gemma-2-9B, we find the operation is carried by a **manipulable operator direction** in the residual stream, largely **separable from the operand**: a two-way analysis of variance factorizes the mid-network state as $H \approx \mu + \text{operand} + \text{operator} + \text{interaction}$, ~90% additive, with the operator component dominant at the query position. The direction is **causal**: adding $v(\text{op_B}) - v(\text{op_A})$ flips the target-vs-source logit margin for **all 20 ordered relation pairs in all three models** (at 1.7B a single mid-workspace layer suffices), while matched-norm random controls do nothing — and a query-position activation patch reroutes the greedy answer itself at the model's competence ceiling. It **generalizes**: directions built on half the operands flip the held-out half; directions built in one wording of a relation transfer to re-framed and fully re-lexicalized prompts; and the operation is separable from its **surface realization** — *language-of* and *demonym-of* are distinct directions even though both emit "Italian". The factorization is **specific to relational retrieval**: under the identical pipeline, arithmetic and comparison-logic operators show 2–4× the interaction and fail held-out generalization. Relational computation in these models is, to a first approximation, an operand plus a transplantable operator.

1 Introduction

When a language model completes "*The currency of Italy is—*", does it represent *currency-of-Italy* as one fused thing, or as an entity plus an operation applied to it? This paper argues, causally, for the second reading.

A line of work shows that a *task* or *relation* can be captured by a single addable vector in an LLM's

residual stream (task vectors, Hendel et al., 2023; function vectors, Todd et al., 2024) and that many relations are approximately linear operators (LRE, Hernandez et al., 2024; Merullo et al., 2024). What has not been characterized is the **joint structure of operator and operand**: whether the operation *factorizes* from its argument, whether that factorization *generalizes*, whether the operation is separable from the *word it produces*, and whether any of this holds *beyond* relational facts, in arithmetic and logic. We answer these on Qwen3 and Gemma-2. A closing discussion (§5) offers a morphological reading — relational computation as **declension** — as an organizing metaphor for the asymmetries we observe.

2 Related work and what is new here

Extracting relational concepts. The closest prior work is Wang et al. (2024), who locate hidden states that express a relation separately from its subject, transplant these relational representations between subjects, and rewrite relations across 22 relation types and several models. We instead characterize the **joint geometry** of relation and entity: we quantify its additive and interaction components (the two-way ANOVA with a measured fusion term), run exhaustive all-pairs swaps against matched-norm nulls, and test whether relation directions generalize causally across held-out entities, prompt contexts, architectures, and task domains — including the domains where the factorization fails.

Operation as an addable vector. Task Vectors (Hendel et al., 2023) and Function Vectors (Todd et al., 2024) extract one vector per ICL task and add/patch it to trigger the task. In-Context Vectors (Liu et al., 2024) and ActAdd (Turner et al., 2023) steer a task/style similarly. These bundle operator *and* operand into one unfactored "task" direction and (for FVs) show *partial* cross-task arith-

metic without null controls. We contribute the **factorization** (operator $\oplus$ operand), an **exhaustive all-pairs swap with matched-norm nulls**, and a **held-out-operand generalization** test.

Relation as a linear operator. LRE (Hernandez et al., 2024) fits a full affine map $\mathbb{W} \cdot s + b$ per relation. That represents the relation as a *monolithic* operator; we show an **additive operator direction** that is arithmetically composable and **factorable from the operand**, and we quantify the operand/operator/interaction variance split, which LRE does not.

Additive factual recall. Summing Up the Facts (Chughtai et al., 2024) decomposes factual recall into additive *circuits* (subject vs relation contributions). Ours is a **representational two-way ANOVA of the residual state** with an explicit, measured **interaction/fusion term** — a different object.

Attribute subspaces / reading position. RAVEL (Huang et al., 2024) disentangles multiple *attributes of one entity* into subspaces (operators sharing an operand); we measure the orthogonal axis, **operator vs operand**. The operand→operator shift along the sequence restates the subject-enrichment → attribute-extraction dynamics of Geva et al. (2023) in operator/operand terms — our least novel point, included as confirmation.

Global workspace and the Jacobian lens. Gurnee et al. (2026) introduce the Jacobian lens and identify a mid-depth "global workspace" band of the residual stream. We adopt their depth band as the site of intervention and their lens as scaffolding; our contribution is orthogonal — the operator/operand structure of what that band *holds* — and we additionally report a controlled null on the lens's readout advantage for our task (§4.8).

Arithmetic and logic. Arithmetic in LLMs is reported as a "bag of heuristics" (Nikankin et al., 2025) and Fourier/helical procedures (Nanda et al., 2023; Zhong et al., 2023; Kantamneni and Tegmark, 2025), with the operand *value* cleanly linear (Gurnee and Tegmark, 2024). Truth is a steerable direction (Marks and Tegmark, 2023) and logical-operator heads exist (Hong et al., 2024). No prior work reports a **cross-domain operator/operand factorization**; we provide one and show it **breaks down** for arithmetic and logic — itself a result.

What is new here is the combination, not any single ingredient. That relations admit addable

vectors (task/function vectors) and linear decoders (LRE) is established; what has not been done is the joint, quantified analysis: (1) an operator $\oplus$ operand **factorization of the residual state itself** (two-way ANOVA with a measured interaction/fusion term and operator/operand subspace angles); (2) an **exhaustive all-pairs causal swap** with matched-norm nulls; (3) **held-out-operand generalization** as the test separating a transferable direction from interpolation among build examples; (4) the **operation-vs-realization dissociation** (language $\neq$ demonym as directions despite identical output); and (5) a **cross-domain test** showing the factorization is specific to relational retrieval and breaks down for arithmetic and logic. The declension/case reading is our *presentation* of these measurements — an organizing metaphor that anticipates the observed asymmetries (syncretism at the exponent, fusion in the interaction term) — not an additional empirical claim.

3 Method

Setup. Qwen3-1.7B/8B throughout, plus Gemma-2-9B for the cross-architecture replication (§4.5); all run in bf16 on a single AMD Strix Halo APU (environment details in the supplementary repository). A relation is rendered into a template; the canonical frame is "The {op} of {a} is", and §4.4 additionally uses two paraphrase frames (question–answer and discourse-prefixed) that hold the {op} of {a} unit fixed while varying the surrounding frame, all ending in *is* so answer scoring is comparable. Reading position is the query token unless stated; the depth "workspace" band (38–92% of layers) is where Gurnee et al. (2026) locates the global workspace and where we build operator directions.

Operator direction. For operator k , $v(k) = \text{mean_operand}[h(\text{operand}, k) - \text{mean_op } h(\text{operand}, \cdot)]$ at the workspace layers — the deviation of an operator from the operand's average over operators, averaged over operands (the function-vector construction; implementation details in Appendix A). Intuitively: what states asking the *same question about different countries* share, once what *all questions about each country* share is removed.

Swap and controls. Efficacy of $k_A \rightarrow k_B$ is the change in $\text{logit}(\text{answer}_B) - \text{logit}(\text{answer}_A)$ at the query position when

$\alpha \cdot (v(k_B) - v(k_A))$ is added over the band, versus a **matched-norm random** direction. Answers are scored by their distinguishing token (bare digit for arithmetic, since Qwen3 splits " 3" into [space, 3]); a tokenization guard drops, within each operator pair, the operands whose two answers collide at that token (for relations this leaves 4 of 12 operands on the syncretic demonym $\leftrightarrow$ language pairs and all 12 elsewhere).

Factorization. $H[\text{operand}, \text{operator}] = \mu + \text{operand}(o) + \text{operator}(k) + \text{interaction}$; we report the variance share of each term and the principal angles between the operand- and operator-subspaces (Appendix A).

Generalization. Build $v(\text{op})$ from half the operands; test the swap on the held-out half. This distinguishes a genuine operator from interpolation among the examples used to build it.

Statistical treatment. The 20 ordered swaps are **not** 20 independent observations: they are built from 5 operator directions (each participating in 8 ordered pairs) and evaluated on the same shared operand set (12 operands, 4 on the syncretic pairs after the tokenization guard). We therefore report cluster-bootstrap percentile intervals (10,000 replicates) at two levels. Within a pair, the **operand** is the resampling unit (per-pair 95% CIs). Across the paradigm, the **operator** is the top-level cluster: a dyadic node bootstrap resamples the operator set with replacement, weights each ordered pair by the product of its endpoints' multiplicities, and resamples operands within surviving pairs (per-operand values are released as long-form artifacts; Appendix A).

J-space controls. We repeat readout-geometry in the J-lens readout $\text{unembed}(J \cdot h)$ vs the logit-lens readout $\text{unembed}(h)$, and re-run efficacy with **spectrum-matched random-projection** and **permuted-vocabulary** null lenses (Appendix B).

4 Results

4.1 Relational operators are manipulable directions

All 20 ordered operator swaps flip the **target-vs-source logit margin** (the sign of $\text{logit}(\text{answer}_B) - \text{logit}(\text{answer}_A)$ at the distinguishing token; mean swap **+21** logit units on 1.7B, **+25** on 8B) while the matched-norm random control is ~ 0 . Representative (1.7B):

swap	clean	+operator	+random
currency $\rightarrow$ capital	-7.9	+31.5	-1.2
capital $\rightarrow$ currency	-12.0	+33.0	-2.5
capital $\rightarrow$ language	-8.5	+32.5	+3.0
continent $\rightarrow$ language	-1.9	+23.0	+1.4

Under the operator-level cluster bootstrap (§3, the level that respects that pairs share directions), the swap-random contrast is **+22.6, 95% CI [+14.0, +32.1]** at 1.7B and **+26.0 [+17.9, +32.8]** at 8B; the flip fraction is **1.00 [1.00, 1.00]** at both scales — every replicate flips every pair. Per-pair operand-bootstrap CIs never cross zero:

Dose-response and collateral cost. Sweeping the intervention strength α (1.7B) separates the effect into its parts: the **operator-specific** component (swap - random) rises and **saturates at $\approx +23$ by the default $\alpha = 4$** , while the matched-norm random control contributes a **nonspecific** margin shift of $\approx +6$ that is flat in α — any large perturbation degrades the clean answer's dominance somewhat, which is why all headline numbers are swap - random contrasts. The intervention is *answer-surgical*, not *distribution-surgical*: with the same hook active on unrelated WikiText, per-token $\text{KL}(\text{clean} \parallel \text{intervened})$ grows from 7.9 nats ($\alpha = 0.5$) to 21 nats ($\alpha = 12$) — 18.4 at the default — so the band-wide edit substantially distorts off-task text. We report this as an honest cost of the band-wide, all-position intervention rather than tuning it away; targeted (single-position) variants are the natural mitigation and are left to future work.

Minimal intervention, and margins vs. generation (1.7B). The band-wide injection is not the minimal effective intervention: restricting the same directions to a **single mid-workspace layer** still flips **20/20 margins** (contrast +6.8 [+3.0, +8.9]) at $4.4\times$ lower off-task cost (KL 4.6 vs. 20.6 nats/token here; the dose sweep's 18.4 at $\alpha = 4$ reflects a different pair/prompt sample); the middle half-band gives +12.8 [+7.9, +16.3] at 10.9 nats. Greedy decoding sharpens the picture: the additive injection — at any width — flips the *relative* margin but does not make the model emit the target answer within three greedy tokens ($\leq 0.5\%$ exact match, 1/224 items at the wider bands, 0/224 at a single layer; consistent with its off-task distortion), whereas **replacing the query-position residual with a real donor activation** (same operand, target relation — classic activation patching: one position, no averaged di-

Operator injection flips the logit margin: 20/20 pairs flip sign (matched-norm random control mean -0.9)

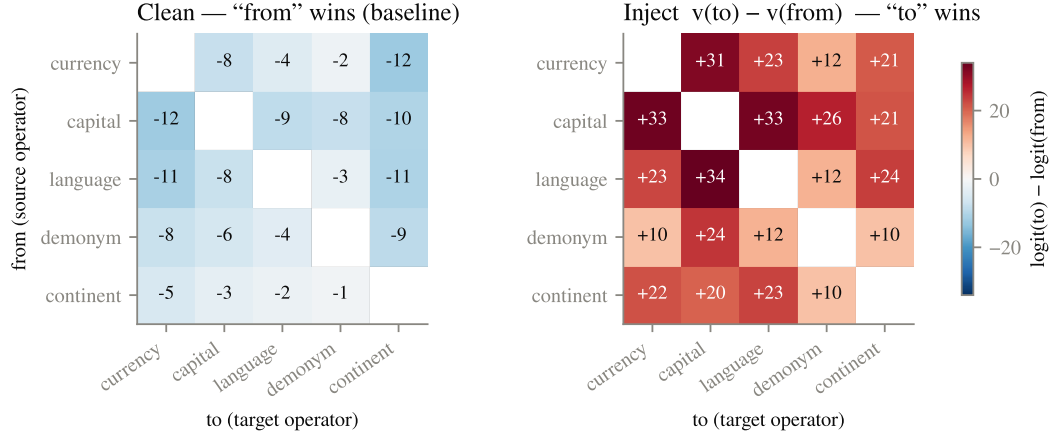


Figure 1: All-pairs operator swap: clean (baseline, "from" wins, blue) vs. injecting $v(\text{to}) - v(\text{from})$ ("to" wins, red). 20/20 pairs flip sign; matched-norm random control ≈ 0 .

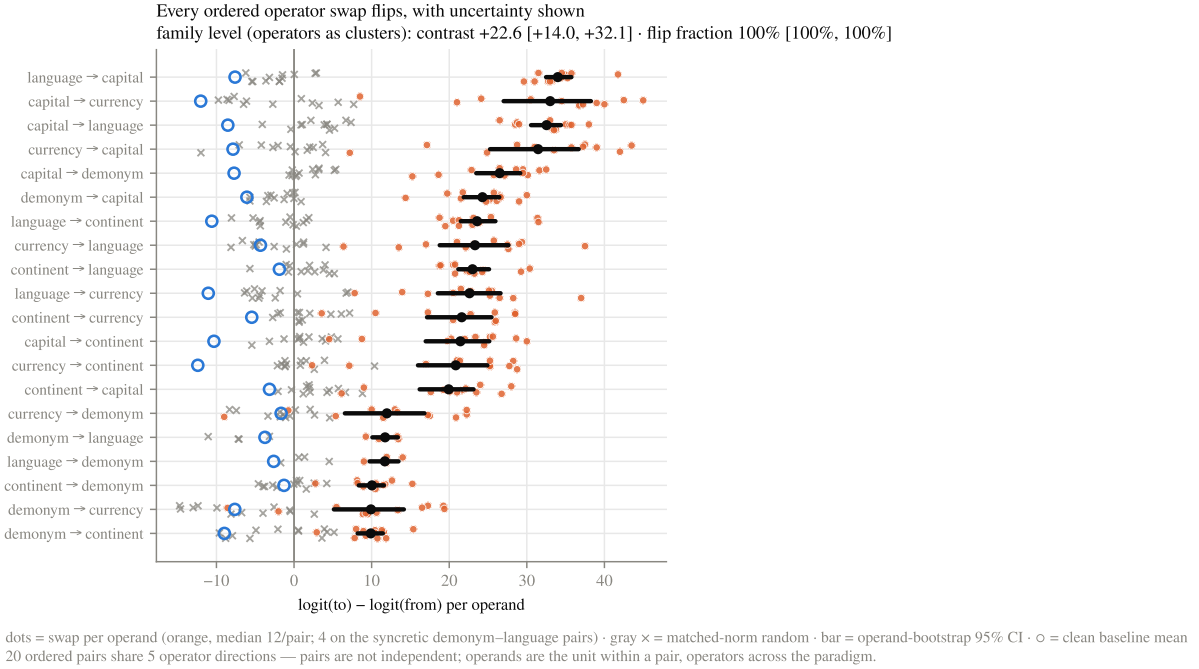


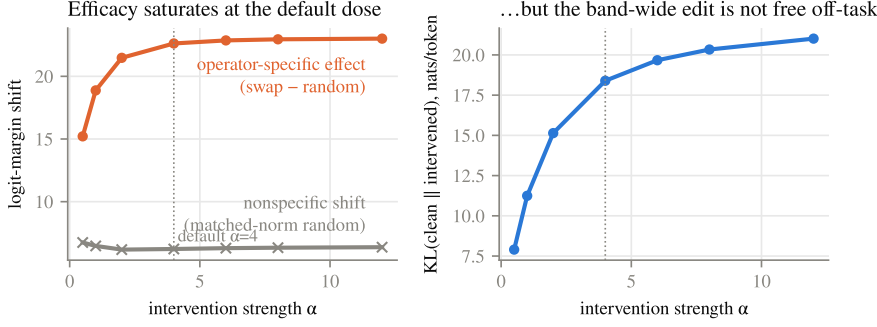
Figure 2: Per-pair distributions: swap values per operand with operand-bootstrap 95% CIs, matched-norm random control, and clean baselines. No CI crosses zero; the weakest pairs are exactly the syncretic ones (demonym↔language and their neighbours).

rection) makes the model *say* the target answer at **51%**, essentially its own clean-prompt accuracy of 53%. The averaged difference direction manipulates preference; the state-level patch reroutes the generated answer at the model's competence ceiling.

4.2 The representation factorizes into operator $\oplus$ operand

Does the state literally decompose into "the thing" plus "the question asked about it"? Two-way ANOVA at a mid-workspace layer (1.7B / 8B):

~90% additive; operand and operator components occupy distinct subspaces (principal angles 41–85° at the query token, all three models — though near-orthogonality is the high-dimensional default, so the load-bearing evidence is the vari-



Right: per-token KL on unrelated WikiText with the hook active at every position of the workspace band — the intervention is answer-surgical, not distribution-surgical.

Figure 3: Dose-response: the operator-specific effect saturates at the default dose $\alpha=4$ while the nonspecific random shift stays flat; per-token KL on unrelated text grows monotonically — the edit is answer-surgical, not distribution-surgical.

read position	operand	operator	interaction (fusion)
query token	5% / 6%	86% / 82%	9% / 13%
entity token	59% / 55%	31% / 34%	9% / 11%

Table 1: Two-way ANOVA of the workspace state: variance shares by read position (1.7B / 8B).

ance decomposition and the causal swaps, not the angles). The operation- dominant representation **emerges along the sequence**: the entity enters operand-dominant and is *declined* into an operator-marked form by the query position (the reading-position control rules out template echo; the causal swaps are the stronger control).

4.3 Operation $\neq$ realization (syncretism)

language-of and *demonym-of* emit the **same word** (Italian) yet are **distinct operator directions**; the swap between them is weak precisely because they share an exponent. A **pure desineness** built from the shared-output pairs ($\text{mean}[\text{h}(\text{language}) - \text{h}(\text{demonym})]$, exponent cancelled) still installs the relation causally (1.7B: $-3.6 \rightarrow +12.5$). The syncretism appears at the *exponent* level, not the *case* level — as in Latin declension.

4.4 Operators generalize to held-out operands and across prompt frames

Held-out operands. $\text{v}(\text{op})$ built on 6 operands and applied to the 6 held-out operands still flips **20/20** swaps — a genuine, transferable operator, not interpolation. Operator-level cluster bootstrap on the held-out contrast: **+20.0 [+10.8, +29.5]** at 1.7B, **+22.8 [+14.0, +31.0]** at 8B; flip fraction **1.00 [1.00, 1.00]** at both.

Cross-frame transfer. To rule out template echo we render every relation in three

paraphrase frames — declarative (The {op} of {a} is), question-answer (Q: What is the {op} of {a}? A: It is), and discourse-prefixed (It is well known that the {op} of {a} is) — and test every build $\rightarrow$ test frame combination: $\text{v}(\text{op})$ built on frame i , all-pairs swap run on frame j . On 1.7B, **all 9 combinations flip 20/20** (180/180; off-diagonal transfer 120/120, mean contrast **+24.2**), and the contrast is *frame-invariant to two decimals*: directions built on the declarative frame produce +22.6 on all three test frames. The clean baselines do shift across frames ($-7.0 / -8.1 / -7.9$), confirming the frames are genuinely different prompts; the operator's causal effect does not. At 8B the pattern replicates — **100/100 flips over the tested combinations (80/80 cross-frame, mean contrast +28.7)**, with the declarative-built direction again frame-invariant ($+26.0 / +26.0 / +26.0$). The operator direction is invariant to the prompt **context** around the fixed {op} of {a} unit.

Cross-lexicalization transfer. The stronger question is whether the direction survives changing the *wording of the relation itself*. We re-lexicalize every relation — *currency of {a}* $\rightarrow$ *money used in {a}*, *capital of {a}* $\rightarrow$ *seat of government in {a}*, *language of {a}* $\rightarrow$ *language primarily spoken in {a}*, *demonym* $\rightarrow$ *name for someone from {a}*, *continent of {a}* $\rightarrow$ *world region*

Where operand and operator live

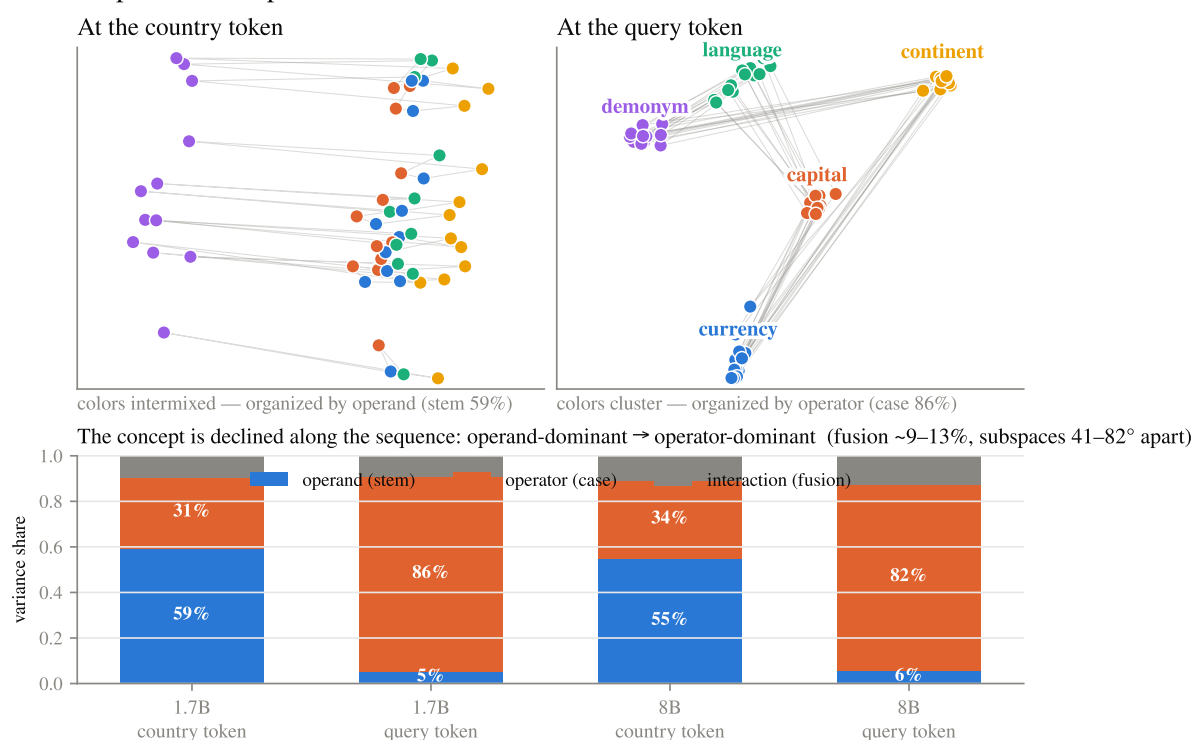


Figure 4: PCA of the 60 workspace vectors $H[\text{operand}, \text{operator}]$, colored by operator. At the country token the cases intermix (operand-organized, stem 59%); at the query token they separate into clean case clusters (operator-organized, case 86%). The faint web links each country's five case-forms — short at the country token, splayed at the query token. Bottom: the variance split, 1.7B and 8B.

Operation $\neq$ realization: the case is separable from the word

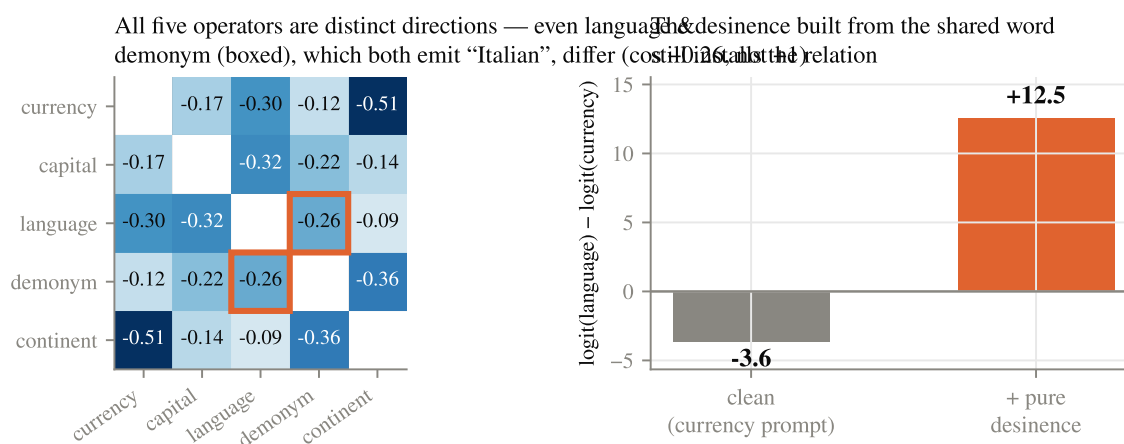


Figure 5: Left: operator-direction cosines — all five operators are distinct directions (no off-diagonal near ± 1); language and demonym, boxed, both emit "Italian" yet differ (cos -0.26). Right: the pure desinence, built where the two share an output word, still installs the relation causally ($-3.6 \rightarrow +12.5$).

containing {a} — and test both transfer directions. Directions built on the of-phrasings flip the re-lexicalized prompts, and vice versa: **40/40 at 1.7B (mean contrast +21.7) and 40/40 at 8B (+26.8)**, with transfer contrasts indistinguishable

from within-formulation ones (1.7B: +22.59 transferred vs. +22.62 within; operator-level 95% CIs overlap almost exactly). The clean baselines differ across formulations — the prompts are genuinely different — but the operator's causal effect does

not. The direction is a property of the **relation**, not of its wording.

4.5 Cross-architecture replication (Gemma-2-9B)

The entire pipeline transfers unchanged to **Gemma-2-9B** — a different pretraining corpus, tokenizer (SentencePiece; answer single-token coverage 0.95), and architecture family (soft-capped logits, GQA):

Every qualitative signature replicates — all-pairs flips with matched-norm nulls ≈ 0 , held-out-operand transfer, case-dominant factorization at the query token with single-digit fusion, the along-sequence shift (stem 37.6% $\rightarrow$ 8.4% from the entity token to the query token), and the exponent-free desinence. Raw logit units are not directly comparable across architectures (Gemma-2 soft-caps its final logits), so cross-model comparison should lean on the flip rates and variance shares, which match or exceed Qwen3's. The operator-operand factorization is not an idiosyncrasy of one model family.

4.6 The factorization is domain-specific (arithmetic and logic do not, under our setup)

If the factorization were an artifact of prompting, it should appear wherever the prompt has the same shape. It does not. Extending the identical pipeline to arithmetic (+, $\times$, -) and comparison-logic operators, on 1.7B:

The gradient replicates and sharpens at **8B**: relational held-out generalization 20/20 (interaction 13%); arithmetic interaction 25%, held-out 1/6; logic interaction 45%, held-out **0/6** (swap below random). Monotone at both scales: relational $\gg$ arithmetic $>$ logic.

Relational operators factorize cleanly and generalize; arithmetic and logical operators are **entangled with their operands** (2–4 $\times$ the interaction) and do **not** form a transferable direction — consistent with arithmetic being a bag of heuristics / Fourier computation rather than a linear operator. The **held-out generalization test is what separates** a transferable operator direction from memorized interpolation.

4.7 Reconciling with a contrary report: add-N vs. two-operand arithmetic

Christ et al. (2025) report the *opposite* sign for arithmetic: an operator built from an **add-N** rela-

tion (a fixed addend N applied to a single number) **does** generalize to held-out relations. The two results do not conflict — they cut arithmetic along different axes. In add-N the operator is **the addend N** over one numeric operand, so "generalizing across N" is interpolation along the number line (the operand *value* is itself linear; Gurnee and Tegmark, 2024), not transfer of a *function*. Our + $\times$ - cut varies the **function** over two operands. Running both cuts on Qwen3 (op_core.py --domain arith_addN vs arithmetic; scripts/operator_collinearity.py) places add-N exactly between relations and the genuine-function cut:

add-N generalizes **better** than + $\times$ - (positive vs. clearly negative) and its operator directions are the **most collinear** of the three families — 76% of the operator-set variance on a single line — consistent with add-N being a linear numeric family rather than a set of distinct operations. This reproduces Christ et al.'s positive (their cut varies a linear parameter) alongside our negative (ours varies the function) with no contradiction. The caveat is quantitative: Qwen3 BPE restricts us to single-digit results, so the add-N grid is small (5 operands) and its generalization is weak in absolute terms — the **ordering** (relations $\gg$ add-N $>$ + $\times$ -), not the magnitude, is the claim.

4.8 This is causal structure, not a J-space readout advantage

This project began as a replication of the J-space readout claim of Gurnee et al., 2026, whose "workspace" band we intervene on throughout. Under matched controls — including a spectrum-matched random projection — the Jacobian-lens readout showed **no advantage** over the logit lens on any of four comparisons (Appendix B). The structure reported here is **causal** organization of the residual stream, not a privileged readable subspace.

5 Discussion: relational computation as declension

The measurements invite a morphological reading, offered as an organizing metaphor rather than an additional claim. In a declining language a noun's role is marked by a case ending on a stem — and each structural feature of a case system has a measured counterpart here. The same stem takes different endings (one country, five oper-

measure	Qwen3-1.7B	Qwen3-8B	Gemma-2-9B
all-pairs swap contrast	+22.6 [+14.0, +32.1]	+26.0 [+17.9, +32.8]	+30.4 [+23.8, +34.8]
flips (swap > 0)	20/20	20/20	20/20
held-out-operand contrast	+20.0 [+10.8, +29.5]	+22.8 [+14.0, +31.0]	+26.6 [+20.6, +33.2]
operator variance @ query	86%	82%	84.8%
interaction (fusion)	9%	13%	6.8%
pure desinence (clean → +v)	-3.6 → +12.5	-2.9 → +8.1	-3.0 → +8.5

Table 2: Cross-architecture replication: every signature reproduces on Gemma-2-9B (logit units are not comparable across architectures; compare flip rates and variance shares).

domain	operator variance	interaction (fusion)	held-out generalization	swap vs random
relational	86%	9%	20/20 flip	+21 vs ~0
arithmetic	55%	23%	2/6 ($\leq$ random)	+0.4 vs +0.3
logic (compare)	33%	34%	2/6 ($\leq$ random)	+1.2 vs +0.2

Table 3: Cross-domain factorization: relational operators factorize and generalize; arithmetic and logic do not, under our setup.

cut	held-out generalization (1.7B / 8B)	operator-set collinearity (top-1 var)
relations (5 operations)	20/20 · 20/20	0.39 (spread paradigm)
add-N (operator = addend)	5/12 · 6/12 (+0.10 / +0.05)	0.76 (most 1-D / number-line)
+ × − (operator = function)	2/6 · 1/6 (-0.43 / -0.55)	0.64

ator directions); the same ending applies across stems (held-out-operand transfer, §4.4); two cases can share a surface form without being the same case (**syncretism**: *language-of* and *demonym-of* both emit "Italian", yet are distinct directions and their mutual swap is the weakest of the paradigm, §4.3); and stem and ending do not compose perfectly (**fusion**: the ~10% interaction term, §4.2). The exponent-free **desinence** is the metaphor's sharpest prediction — a case marker stripped of its surface form should still be causally functional, and it is (§4.3). The metaphor also marks its own limits: arithmetic and logic operators do not decline (§4.6); whatever those domains use, it is not a case system.

6 Limitations

- **Scale and family.** 1.7–9B models from two families (Qwen3, Gemma 2) — both decoder-only transformers; no ≥ 30 B model. Relational linearity holds for a subset of relations even in-domain (cf. LRE ~48%).
- **Relation coverage.** Five hand-chosen country relations, English-only. Prompt variation covers three context frames and a full relexicalization of every relation (§4.4); more relations, other entity types, and other languages remain open.
- **Arithmetic coverage.** Qwen3 BPE splits

multi-digit numbers, so we restrict to single-digit results; the arithmetic null is on that regime and at 1.7B/8B, where arithmetic competence is itself limited.

- **Reading-position result** restates known subject-enrichment→attribute-extraction dynamics.
- **No J-space claim.** The readout-geometry difference is 1.7B-specific; the paper stands on the causal factorization, not on a J-lens readout advantage.

7 Conclusion

An LLM represents a relational operation as a causally manipulable operator direction that factorizes from the operand, generalizes to unseen operands, and separates the operation from the word that realizes it — a structure well described as **declension**. This factorization is not universal: it is clean for relational retrieval and breaks down for arithmetic and logic, whose operators are entangled with their operands. The structure is causal; it is not a more-readable subspace.

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A Reproducibility, stimuli, and implementation details

Checkpoints and hardware.

Qwen/Qwen3-1.7B, Qwen/Qwen3-8B, google/gemma-2-9b, all bf16 on a single AMD Strix Halo APU (128 GB unified memory, ROCm PyTorch). Seeds default to 0 and affect only the random controls; direction building is deterministic.

Stimuli. Operands (12 countries): Italy, Japan, Russia, China, France, Germany, Brazil, Egypt, India, Mexico, Sweden, Turkey. Relations: currency, capital, language, demonym ("word for a person from"), continent; every (country, relation) cell has a gold one-word answer (e.g. Italy → euro, Rome, Italian, Italian, Europe); the full grid ships in `data/relations.json`, the lexical reformulations in `data/relations_lex.json`, and the paraphrase frames in the dataset metadata.

Answer scoring and selection. Answers are scored at the first token of " " + answer. Within each ordered operator pair, operands

whose two answers collide at that token are dropped (4 of 12 survive on the syncretic demonym↔language pairs; all 12 elsewhere). Whole-word single-token coverage: 0.92 (Qwen3 BPE), 0.95 (Gemma SentencePiece).

Directions, interventions, and controls. $v(\text{op})[\text{layer}]$ is the mean over (operand, frame) cells of the op's residual minus that cell's mean over ops, computed at ~25 evenly spaced source layers restricted to the 38–92% depth band. Interventions add $\alpha \cdot (v(\text{to}) - v(\text{from}))$ at every position of the band ($\alpha = 4$ unless swept). The matched-norm control draws one Gaussian vector per layer and rescales the band-concatenated stack to exactly the injected difference's norm. Clean mean margins $\text{logit}(\text{to}) - \text{logit}(\text{from})$ are ≈ -7 to -8 across models and frames; per-operand values for every experiment are released as long-form artifacts (results/ablation/*_long.parquet).

Experiment drivers. op_core.py (directions, swaps, ANOVA, cluster bootstraps), operator_paradigm.py, operator_factorize.py, operator_templates.py, operator_lexical.py, op_dose.py, op_minimal.py, op_geometry_dump.py.

B The J-space readout comparisons

Four readout comparisons on Qwen3-1.7B (8B where noted) asked whether the Jacobian-lens readout exposes more than the logit lens: bridge-entity pass@k (a tie, drifting toward the logit lens at 8B); a surface-form logit difference with real competitors (logit lens matches or wins in every depth band); a grammatical-number probe with Hewitt–Liang selectivity (number decodable at ceiling from *all* lenses — including a **spectrum-matched random projection**, the decisive control); and the concept-plane trajectory that motivated the project (the J-lens path is longer on random planes too — a generic magnitude effect, not a concept channel). A permuted-vocabulary lens control behaves accordingly. The 1.7B readout-geometry difference (cases reorganizing by output form in J-space) does not replicate at 8B and is not relied on. Full tables live in the project's working log.